Project Executable Files

Date	01 July 2026
Team ID	xxxxxx
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section provides the executable files, source code, and supporting resources required to run the OptiCrop application. The project is organized into modular directories containing the machine learning model, data preprocessing scripts, Flask web application, testing files, and documentation. All required dependencies and execution instructions are included to enable successful project setup and execution**.**

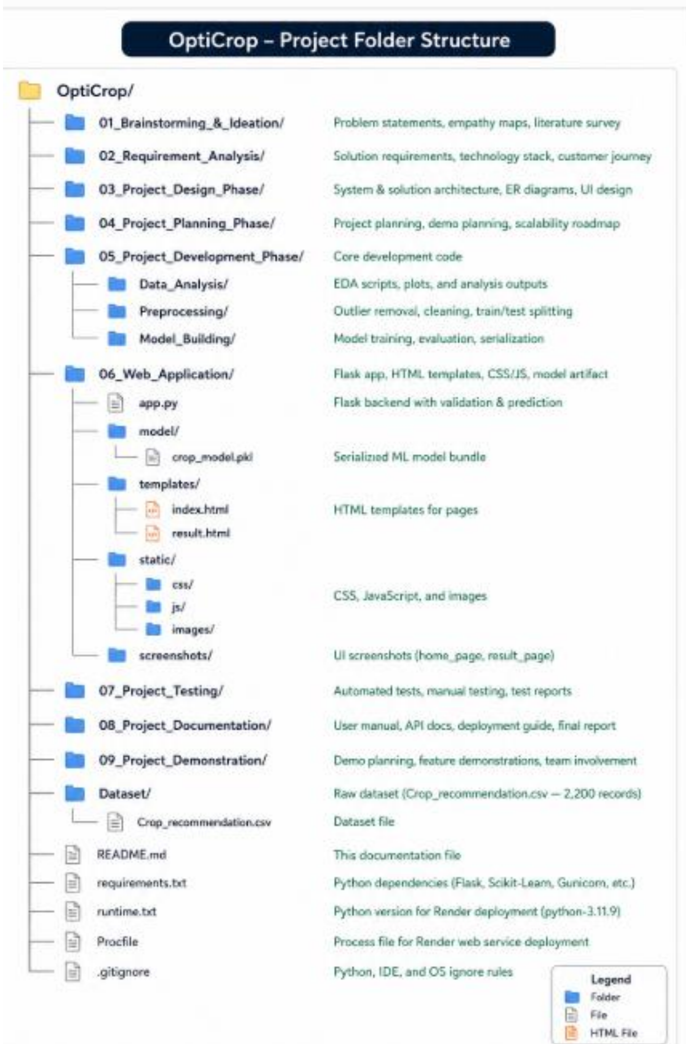
Step 1: Submission Checklist

S.No	**Item to Submit**	**Submitted (Yes / No / NA)**
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	Yes
5	Environment configuration file (.env.example or similar)	No
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	No
9	Test files and test results	Yes

10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:



Step 3: Deployment / Access Details

Field	**Details**
Hosted / Deployed URL	https://optic-crop.onrender.com/
Login Credentials (Demo)	Not Required (Public Access)

Platform / Hosting Provider	Render
Repository Link	https://github.com/sivaganesh7/Optic-Crop
Demo Video Link	https://drive.google.com/file/d/11YvnjCxxBHIX8V81ddLnw2Esp5jnu-zG/view?usp=drivesdk

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

1. Clone the repository from GitHub.
2. Open a terminal and navigate to the project directory.
3. Create a virtual environment using:
   `python -m venv venv`
4. Activate the virtual environment:
   - Windows: `venv\Scripts\activate`
5. Install project dependencies:
   `pip install -r requirements.txt`
6. Run the application:
   `python run.py`
7. Open the provided local URL in a web browser.
8. Enter soil and environmental parameters and submit the form to receive crop recommendations.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Recommendations depend on the quality of input data	Ensure accurate soil and environmental values are entered
2	Real-time weather integration is not available	Future enhancement planned
3	Mobile application support is not currently available	Planned for future development

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Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1		
2		
3		